MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026 : SECOND SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCAQC100
INTRODUCTION TO AQUARIUM AND ORNAMENTAL FISHES
(2024 ADMISSION ONWARDS) : **Q paper Code : 25901598**

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C),Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1. What is the main difference between freshwater and marine aquaria? [U] / [CO1]

Salinity of the water

Freshwater Aquarium:Contains water with little to no salt, Houses fish like guppies, goldfish, and tetras, Easier and cheaper to set up and maintain, Water chemistry is simpler and more stable.

Marine Aquarium:Contains saltwater with carefully controlled salinity, Houses marine organisms like clownfish, corals, and starfish, More complex and expensive due to need for salt mix, protein skimmers, and precise monitoring, Requires strict control of water parameters (salinity, pH, calcium, etc.).

2. What type of aquarium is specifically designed for marine species like corals and saltwater fish? [U] / [CO1] : Marine aquarium: Reef aquarium.

3. What is the purpose of tunnel walkways in Oceanariums? [U] / [CO1]

To give visitors an immersive, underwater viewing experience.Observe marine animals (sharks, rays, fish) from all sides—above, around, and in front, Feel as if they are walking under the sea, View natural swimming behavior without disturbing the animals.

4. Explain the importance of regular cleaning and maintenance of an aquarium. [U] / [CO1]

To keep the aquatic environment healthy, stable, and safe for its inhabitants, it helps to maintain good water quality, reduce the effect of ammomia, nitrite and nitrate, which otherwise can stress or kill fish, maintenance ensures the health of fish and plants, it prevents algae overgrowth. Maintenance keeps equipment functioning properly, regular care improves the aesthetic appeal of the aquarium.

5. How can we address the rise in ammonia levels? [U] / [CO2]

Rising ammonia levels in an aquarium are dangerous because ammonia is highly toxic to fish. perform a partial water change (25–50%), improve biological filtration by supporting the Nitrogen Cycle, Ensure beneficial bacteria are well established in the filter media, Avoid over-cleaning the filter, which can remove these bacteria, reduce

feeding and remove uneaten food, avoid overstocking, clean the substrate and remove decaying organic matter like dead plants or fish, Add ammonia-neutralizing conditioners , Increase aeration

6. Write the scientific names of two marine ornamental lobsters of India. [K] / [CO3]

*Panulirus ornatus, Panulirus versicolor, P. Longipes, P.polyphagus (*Mention any two)

7. What types of marine life can we include in an aquarium? [U] / [CO2]

A marine aquarium can include a wide variety of marine life, depending on the size of the tank and the level of care provided. 1. Marine fishes such as clownfish, angelfish, butterflyfish, and tangs 2. Corals 3. Crustaceans such as Shrimps (cleaner shrimp), crabs, and lobsters 4. Molluscs such as Snails, clams, and octopuses 5. Echinoderms such as Starfish and sea urchins 6. Other invertebrates such as Sea anemones, sponges, and worms

8. Write the scientific names of any two crustacean copepod parasites infecting fishes. [K] / [CO4] : *Lernaea cyprinacea, Caligus elongatus , Lepeophtheirus salmonis , Ergasilus sieboldi* (any two)
9. Name the aquatic crustacean also known as brine shrimp. [K] / [CO4] : *Artemia salina*
10. Mention the scientific name of the Guppy fish. [K] / [CO5]: *Poecilia reticulata*
11. What role does LED lighting play in modern aquariums? [U] / [CO1]

LED light provides efficient, customizable, and biologically supportive illumination, proper light supports processes like Photosynthesis, helping plants and symbiotic algae in corals grow and stay healthy, LED lights are energy-efficient and long-lasting, they allow customization and control, reduce stress and supports natural behavior in fish. LEDs enhance the visual appeal of the aquarium

12. How do floating plants differ from submerged plants? [U] / [CO5]

Floating plants and submerged plants differ mainly in their position in water, structure, and function.

Floating plants: Float on the surface of the water, Roots may hang freely in water or may be reduced, Leaves are exposed to air and receive direct sunlight, Help in shading the aquarium, reducing algae, and absorbing excess nutrients , Examples : duckweed and water lettuce

Submerged plants: Grow completely under water, Roots are usually anchored in the substrate, Leaves are adapted to live underwater (thin, flexible), Play a major role in oxygen production through Photosynthesis and provide shelter for fish , Examples : Hydrilla and Vallisneria

Part B
Short Answer Questions
Answer any 3 questions

Each question carries 5 marks

13. List and explain the benefits of developing the aquarium fish industry as a cottage industry. [K] / [CO1]

Cottage industry (small-scale, home-based enterprise) offers several economic, social, and environmental benefits:

Explain points such as Low investment and easy setup, Employment generation, Quick returns and regular income, Export potential, Utilization of small spaces, Promotes entrepreneurship, Supports allied industries, Conservation awareness, Promotes captive breeding of ornamental species, reducing pressure on wild populations and encouraging sustainable practices.

14. Explain the different methods for controlling snail populations in an aquarium. [U] / [CO2]

Controlling snail populations in an aquarium is important because excessive snails can spoil aesthetics, compete for food, and indicate poor tank management. Effective control involves a combination of methods: Explain Manual removal, Control feeding, Maintain cleanliness, Biological control, Chemical control (with caution), Preventive measures such as Quarantine and rinse new plants before adding them, as snail eggs often hitchhike on plants , Inspect decorations and substrate before introduction

15.Explain the different types of packaging materials used for transporting ornamental fishes. [A] / [CO3, CO5]

Transporting ornamental fishes requires packaging that ensures oxygen supply, temperature stability, and minimal stress. Explain 1. Polythene bags (plastic bags), 2. Thermocol (polystyrene) boxes, 3. Corrugated cardboard boxes, 4. Oxygen cylinders and accessories, 5. Plastic containers or jars, 6. Glass aquaria (temporary transport tanks), 7. Breather bags (advanced packaging), 8. Cushioning materials

16. Discuss the importance of protein in artificial feed for ornamental fishes. [K] / [CO4]

Protein is the most important nutrient in artificial feed for ornamental fishes because it directly supports growth, health, and overall performance. Explain points such as 1. Growth and tissue development, 2. Repair and maintenance, 3. Reproduction, 4. Color enhancement, 5. Energy source, 6. Immunity and disease resistance, 7. Feed efficiency

17. Explain the various categories of oviparous ornamental fishes. [K] / [CO5]

Oviparous ornamental fishes are those that lay eggs, and based on their breeding behavior and parental care, they are classified into several categories:

Explain with examples 1. Egg scatterers, 2. Egg depositors, 3. Bubble nest builders, 4. Mouth brooders, 5. Nest builders (non-bubble)

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